

Adaptive State-of-Health Forecasting for Second-Life LFP Cells Using Physics-Informed Synthetic Data and Hybrid Model Selection

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Second-life reuse of retired lithium iron phosphate (LFP) cells requires fast prediction of future degradation for each incoming cell. Instead of assuming a single degradation model is optimal for all cells, we develop an adaptive forecasting framework that selects between simple empirical extrapolation and a PyBaMM-trained neural operator based on early-cycle degradation complexity.

Approach (Figure 1). A short diagnostic characterisation and the first $K=50$ operating cycles provide early-cycle context. A Neural ODE [3] is trained on PyBaMM [1] SPMe degradation trajectories generated around seven anchor cells from three LFP suppliers (CALB, EVE, REPT) on Prada2013 chemistry [2]. In parallel, an exponential model is fitted to the same context. A context-holdout residual, computed by fitting the first 80% of the context and predicting the final 20%, selects the neural operator when the residual exceeds a threshold chosen using synthetic trajectories only.

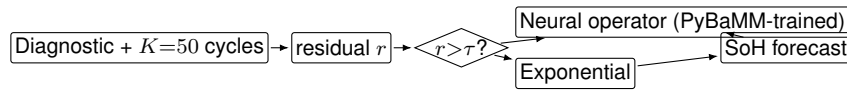


Figure 1: Adaptive selector: a context-holdout residual (exp fit on first 80% of the $K=50$ context, RMSE on last 20%) picks the neural operator above threshold τ ; τ is set on synthetic trajectories only.

Results. Across three external LFP cells excluded from operator training, exponential forecasting achieved 0.70 pp mean RMSE and the PyBaMM-trained neural operator achieved 0.80 pp. The residual-based selector combined their complementary strengths, reducing mean RMSE to **0.47 pp** — a 33% reduction relative to the best fixed strategy.

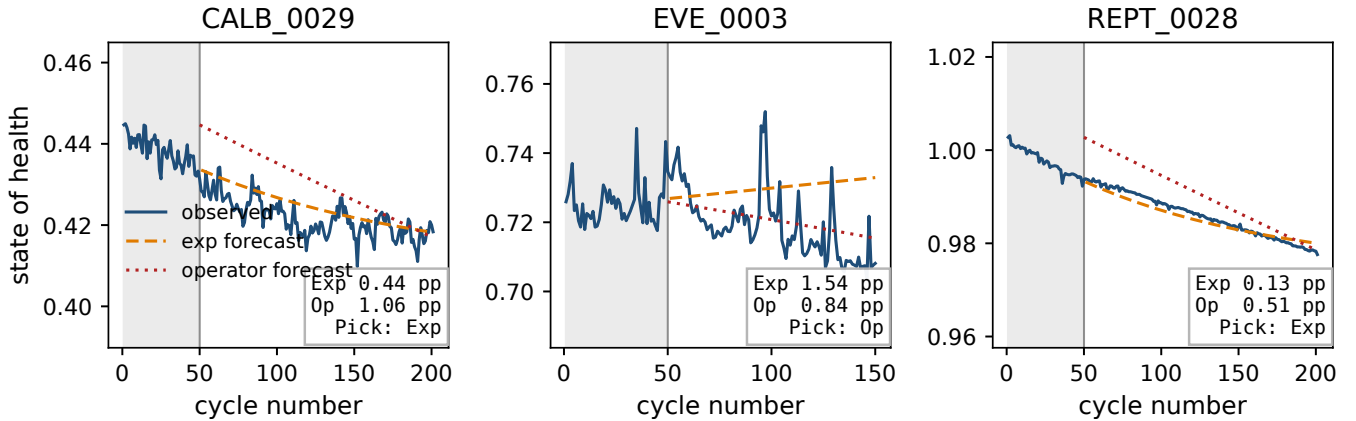


Figure 2: Observed and forecast SoH for the three external cells. Grey band = $K=50$ context; orange dashed = exponential forecast; red dotted = neural operator forecast. Panels use independent y-axis ranges for readability. The context-holdout residual selects the exponential model for CALB_0029 and REPT_0028 and the operator for EVE_0003 — matching the per-cell oracle choice.

Matched ablations indicate that PyBaMM-generated trajectories provide useful training diversity, while explicit degradation-parameter conditioning adds little additional benefit (0.80, 0.82 and 0.90 pp mean RMSE for full-, no- and reduced- θ operators).

The current evaluation is limited to three external cells, and longer-horizon projections require additional experimental validation.

[1] V. Sulzer et al., *J. Open Res. Software* **9** (2021), 14. [2] M. Prada et al., *J. Power Sources* **219** (2012), 353–362.

[3] R. T. Q. Chen et al., *NeurIPS* (2018).